

WAAM Digital Twin — Formula Reference

Illustrated guide aligned with `waam_twin`

waam_twin project

October 6, 2026

Abstract

This document explains every major equation used by the WAAM (Wire Arc Additive Manufacturing) digital twin — not only the symbols, but *what they mean, why they appear, and how the current `waam_twin` code uses them*.

The twin is a GPU continuum melt-pool simulator (Taichi): lattice Boltzmann flow (D3Q19), enthalpy–porosity phase change, VOF free surface, and a catalogue of weld-pool forces. Equations below are grouped by physics domain and matched to `solvers/coupled_step.py`, `kernels/`, and `physics/` as of the current repository (Gaussian / Goldak heat sources, Lin–Eagar arc pressure, energy-normalized injection, physics tiers `thermal/flow/full`).

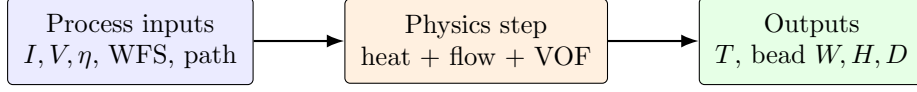
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1 How to read this document

Think of the twin as answering: “Given current, voltage, wire feed, and torch motion, what does the melt pool and bead look like?”



- Each subsection starts with a short **plain-words** explanation.
- Then the equation(s), then a short note on **where the code does it**.
- TikZ sketches show geometry / energy flow; they are conceptual, not CAD.

Notation habit. Capital Q is usually a *power* [W]. Lower-case q is usually a *flux or density* [W/m² or W/m³], or line energy [J/m]. Subscripts: eff = after efficiency, arc = arc source field.

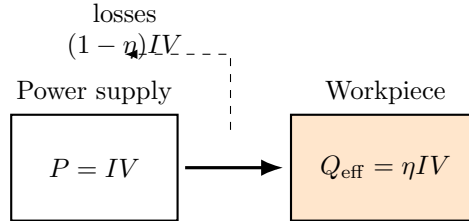
2 Process inputs and energy balance

2.1 Electrical arc power and effective heat

In plain words: The welding power supply pushes current I through an arc with voltage V . Their product IV is the electrical power. Only a fraction η of that power actually heats the workpiece — the rest is lost to radiation, the arc column, spatter, etc. That useful part is Q_{eff} .

$$P = IV \quad (1)$$

$$Q_{\text{eff}} = \eta IV \quad (2)$$



Typical GMAW steel: $\eta \sim 0.6\text{--}0.85$. Your example jobs often use $\eta \approx 0.72$ (`process.arc_efficiency`).

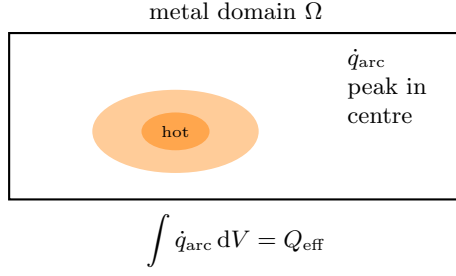
In the code: `twin.Q_w` stores electrical power IV [W]. Efficiency `twin.eta` is applied in the heat kernels so exactly $\eta Q_w \Delta t$ joules enter metal each step (`kernels/thermal.py: inject_arc_heat, inject_goldak_heat`). Do *not* bake η into `Q_w` twice.

References: [24, 20, 34]

2.2 How Q_{eff} relates to $\dot{q}_{\text{arc}}$

In plain words: Q_{eff} is one number: “how many watts of useful heat?” $\dot{q}_{\text{arc}}(x, y, z)$ is a 3D map: “how many watts per cubic metre at each point?” The map is shaped by Gaussian or Goldak. If you integrate the map over the whole domain you must recover Q_{eff} .

$$\int_{\Omega} \dot{q}_{\text{arc}}(\mathbf{x}) dV = Q_{\text{eff}} = \eta IV \quad (3)$$



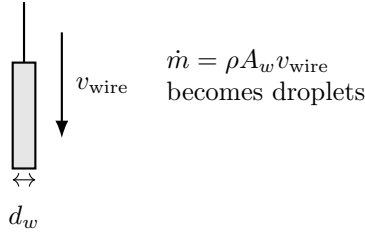
In the energy equation, $\dot{q}_{\text{arc}}$ is the source term (Section 3.3).

2.3 Wire mass feed

In plain words: Wire is pushed into the arc at speed v_{wire} . The cross-section is a circle of diameter d_w . Mass per second is density $\times$ area $\times$ speed.

Wire area $A_w = \pi d_w^2 / 4$:

$$\dot{m} = \rho A_w v_{\text{wire}} = \rho \frac{\pi d_w^2}{4} v_{\text{wire}} \quad (4)$$



In the code: Job fields: `process.wire_feed_m_min`, `process.wire_diameter_mm`. Converted to SI in the twin; used by deposition / droplet sizing.

References: [41, 27, 3]

2.4 Droplet mass and volume

In plain words: Metal does not arrive as a continuous hose of liquid in this model — it arrives as discrete droplets at frequency f_{drop} . Each droplet carries mass $\dot{m} / f_{\text{drop}}$.

$$m_{\text{drop}} = \dot{m} \Delta t_{\text{drop}} = \frac{\dot{m}}{f_{\text{drop}}} \quad (5)$$

$$V_{\text{drop}} = \frac{m_{\text{drop}}}{\rho} \quad (6)$$

Equivalent spherical radius in grid cells (volume V , cell size Δx):

$$r_{\text{cells}} = \left(\frac{3V}{4\pi \Delta x^3} \right)^{1/3} \quad (7)$$

References: [41, 27, 3]

2.5 Travel speed and line energy

In plain words: If the torch moves faster, the same watts are spread over more weld length, so each millimetre of bead gets less energy. Line energy q_{line} captures that.

$$q_{\text{line}} = \frac{Q_{\text{eff}}}{v_{\text{travel}}} \quad [\text{J/m}] \quad (8)$$

References: [36, 15, 24]

3 Governing conservation equations

3.1 Continuity

In plain words: Mass is conserved: fluid cannot appear or disappear except where we explicitly deposit wire. For nearly incompressible liquid metal, density is almost constant ρ_0 .

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (9)$$

In the incompressible LBM liquid region, $\rho \approx \rho_0$.

References: [5, 25]

3.2 Momentum (Navier–Stokes)

In plain words: Newton’s second law for a fluid: acceleration equals pressure forces + viscous drag + gravity/body forces. All the weld-pool mechanisms (Marangoni, Lorentz, arc pressure, ...) enter through $\mathbf{F}_{\text{body}}$.

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \nabla \cdot [\mu(\nabla \mathbf{u} + \nabla \mathbf{u}^T)] + \rho \mathbf{g} + \mathbf{F}_{\text{body}} \quad (10)$$

References: [30, 24, 20]

3.3 Energy / enthalpy

In plain words: Instead of storing temperature alone, the twin stores volumetric enthalpy H [J/m³]. That makes melting/freezing easier: latent heat is already inside H . Heat moves by advection (carried with flow) and conduction; the arc adds $\dot{q}_{\text{arc}}$.

$$\frac{\partial H}{\partial t} + \mathbf{u} \cdot \nabla H = \nabla \cdot (k \nabla T) + \dot{q}_{\text{arc}} \quad (11)$$

with $k = \rho c_p \alpha$ and T recovered from H (Section 5).

References: [40, 20, 7]

4 Heat sources

Valid job values today: `gaussian2d` or `goldak` (aliases `goldak3d`, `doubleellipsoid`). Set via `heat_source` in the job YAML. `conical` is *not* implemented.

4.1 Gaussian surface / near-surface source

In plain words: Imagine a bell curve of heat under the torch: hottest at the centre, weaker farther away. Radius σ sets how wide the bell is.

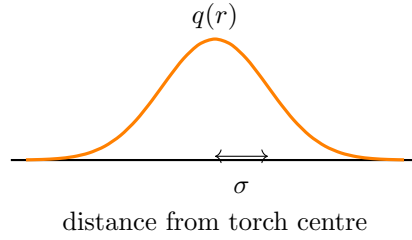
Analytic surface flux idea:

$$q(r) = \frac{Q_{\text{eff}}}{2\pi\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) \quad (12)$$

What the twin actually does (discrete, energy-safe): two passes over metal cells — (1) sum Gaussian $\times$ deposition weights w ; (2) deposit

$$\Delta H_{ijk} = \frac{\eta Q_w \Delta t}{(\sum w) \Delta x^3} w_{ijk} \quad (13)$$

so *exactly* $\eta Q_w \Delta t$ joules enter metal every step, even if the analytic footprint overhangs gas or the plate edge.



In the code: `inject_arc_heat`; arc radius from `arc_physics.sigma_mm` ($\rightarrow$ `twin.sigma_cells`). Optional depth weighting via `penetration_mm` / `arc_surface_weighting`.

References: [37, 15, 3]

4.2 Goldak double-ellipsoid source

In plain words: Real arcs dump more heat behind the torch than in front (the pool trails the motion). Goldak models this with two half-ellipsoids: a short, intense *front* and a longer *rear*, with heat fractions f_f and f_r .

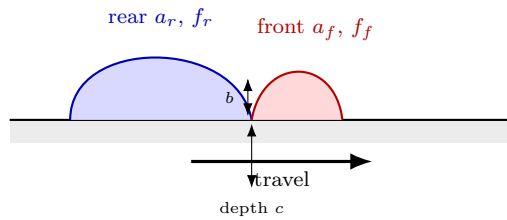
Classical Goldak (1984) power density (front or rear):

$$q_{f,r}(\xi, y, z) = \frac{6\sqrt{3} f_{f,r} Q_{\text{eff}}}{\pi \sqrt{\pi} a_{f,r} b c} \exp\left(-3\frac{\xi^2}{a_{f,r}^2} - 3\frac{y^2}{b^2} - 3\frac{z^2}{c^2}\right) \quad (14)$$

with the Goldak convention

$$f_f + f_r = 2 \quad (15)$$

(not 1 — the factor 2 is part of the classical normalisation). Semi-axes: a along travel, b transverse half-width, c depth.



Typical defaults in jobs: $f_f = 0.6$, $f_r = 1.4$ (sum = 2). YAML axes (mm) map to cells via Δx : `a_front_mm`, `a_rear_mm`, `b_mm`, `c_mm`.

Implementation notes (current code):

- Spatial weight follows the Goldak exponential $f \exp(-3\xi^2/a^2 - 3y^2/b^2 - 3z^2/c^2)$; the closed-form amplitude $6\sqrt{3}/(\pi\sqrt{\pi abc})$ is absorbed by discrete energy renormalisation (same two-pass idea as Gaussian).
- Front/rear is chosen in the *travel frame*: $\xi = (x - x_0)d_x + (y - y_0)d_y$ with unit torch direction (d_x, d_y) from the path — not fixed to grid $+x$.
- Loader enforces $f_f + f_r = 2$ (`normalize_goldak_fractions`).

In the code: `physics/arc.py` (Goldak3D) and `inject_goldak_heat`. See also `docs/PHYSICS_FOR_CE_CORRECTNESS_SPEC.md` §5.4.

References: [15, 16, 3]

4.3 Arc deposition / penetration weight

In plain words: Heat should land mostly near the free surface of the metal, not deep inside solid plate. An optional exponential weight fades heat with depth below the arc.

$$w_{\text{dep}}(z) = \exp\left(-\frac{\max(0, z_{\text{arc}} - z)}{\delta}\right) \quad (16)$$

In the code: `arc_deposition_weight`; `arc_physics.penetration_mm`, `arc_surface_weighting` / `simulation.arc_surface_weighting`.

References: [15, 36, 3]

4.4 Rosenthal far-field check (validation only)

In plain words: Rosenthal treats the arc as a moving point source. It is useful far from the pool for cooling-rate checks, but *not* used as the production heat source in jobs.

$$\sigma_R = \frac{\rho c_p v}{2k} \quad (17)$$

$$T - T_0 = \frac{\eta Q}{2\pi k} \frac{1}{|x|} \exp(\sigma_R x), \quad x < 0 \quad (18)$$

In the code: `validation/test_rosenthal_farfield.py`.

References: [36, 3]

5 Thermal transport and phase change

5.1 Fourier conduction

In plain words: Heat flows from hot to cold, proportional to the temperature gradient. Thermal diffusivity $\alpha = k/(\rho c_p)$ says how fast temperature spreads.

$$\mathbf{q} = -k \nabla T, \quad \alpha = \frac{k}{\rho c_p} \quad (19)$$

References: [20, 9]

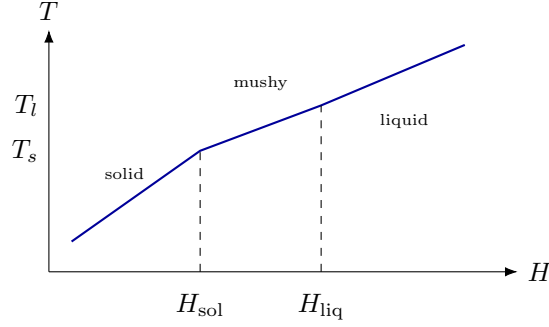
5.2 Enthalpy–porosity method

In plain words: Solid, mushy, and liquid are encoded in H . Below solidus: all solid. Between solidus and liquidus: mixture (liquid fraction f_l). Above liquidus: all liquid, and extra enthalpy raises temperature further.

Define

$$H_{\text{sol}} = \rho c_p T_s \quad (20)$$

$$H_{\text{liq}} = H_{\text{sol}} + \rho L_f \quad (21)$$



Solid ($H \leq H_{\text{sol}}$):

$$f_l = 0, \quad T = \frac{H}{\rho c_p} \quad (22)$$

Mushy ($H_{\text{sol}} < H < H_{\text{liq}}$):

$$f_l = \frac{H - H_{\text{sol}}}{\rho L_f}, \quad T = T_s + f_l (T_l - T_s) \quad (23)$$

Liquid ($H \geq H_{\text{liq}}$):

$$f_l = 1, \quad T = T_l + \frac{H - H_{\text{liq}}}{\rho c_p} \quad (24)$$

In the code: `kernels.update_phase` / thermal phase helpers.

References: [40, 7, 3]

5.3 Lattice thermal update

In lattice units ($\alpha_{\text{lu}} = \alpha \Delta t / \Delta x^2$):

$$\left. \frac{\partial H}{\partial t} \right|_{\text{cell}} = -\mathbf{u} \cdot \nabla H + \rho c_p \alpha_{\text{lu}} \nabla^2 T \quad (25)$$

(first-order upwind advection, central diffusion; variable- c_p path when material tables are on).

References: [20, 31, 3]

5.4 Enthalpy ceiling (vaporization cap)

In plain words: Without a cap, a tiny cell under the arc can spike to absurd temperatures. A vapor enthalpy ceiling clips H near boiling / vaporisation.

$$H \leq H_{\text{vap}} \approx \rho c_p T_{\text{vap, cap}} + \rho L_{\text{vap}} \quad (26)$$

(gated by `enable_enthalpy_cap`, temperature from `T_vapor_cap_K`).

References: [28, 3]

5.5 Evaporative cooling (surface sink)

In plain words: When the surface approaches boiling, evaporation removes enthalpy (energy leaves with vapour). This is separate from the hard ceiling and from recoil pressure.

Phenomenological sink on interface cells for T above an onset toward T_b , scaled by job key `evap_cooling_scale` (under `advanced_physics`) and L_v .

In the code: `apply_evaporative_enthalpy_sink`; flag `enable_evaporative_cooling`. Onset shared with Clausius–Clapeyron recoil.

References: [28, 38, 3]

5.6 Stefan solidification benchmark

$$\text{Ste} = \frac{c_p (T_{\text{init}} - T_0)}{L_f} \quad (27)$$

$$\sqrt{\pi} \lambda \exp(\lambda^2) \text{erf}(\lambda) = \text{Ste} \quad (28)$$

$$s(t) = 2\lambda\sqrt{\alpha t} \quad (29)$$

In the code: `validation/test_stefan_solidification.py`.

References: [9, 11, 3]

5.7 Boundary heat losses

In plain words: The bead and plate lose heat to air by convection and radiation.

On gas-exposed faces:

$$q_{\text{loss}} = h_{\text{conv}} (T - T_{\text{amb}}) + \varepsilon \sigma_{\text{SB}} (T^4 - T_{\text{amb}}^4) \quad (30)$$

$$H \leftarrow H - q_{\text{loss}} \frac{\Delta t}{\Delta x} \quad (31)$$

(H is volumetric; dividing by Δx converts a surface flux to a cell update.)

In the code: `enable_heat_loss`, `heat_loss.h_conv`, radiation flag.

References: [20, 24, 3]

6 Lattice Boltzmann numerics

6.1 Lattice units

In plain words: Physical metres and seconds are converted to dimensionless lattice units so the LBM stays stable (small Mach number).

$$\Delta t = \Delta x \frac{u_{\text{ref,lu}}}{u_{\text{ref,phys}}} \quad (32)$$

$$\nu_{\text{lu}} = \frac{\mu}{\rho} \frac{\Delta t}{\Delta x^2}, \quad \tau = 3\nu_{\text{lu}} + \frac{1}{2} \quad (33)$$

$$\alpha_{\text{lu}} = \alpha \frac{\Delta t}{\Delta x^2}, \quad \tau_T = 3\alpha_{\text{lu}} + \frac{1}{2} \quad (34)$$

Target Mach $\text{Ma} \approx u_{\text{ref,phys}} \Delta t / \Delta x \sim 0.05$.

References: [25, 10, 3]

6.2 D3Q19 equilibrium (SRT)

$$f_i^{\text{eq}} = w_i \rho \left(1 + 3 \mathbf{e}_i \cdot \mathbf{u} + \frac{9}{2} (\mathbf{e}_i \cdot \mathbf{u})^2 - \frac{3}{2} \mathbf{u}^2 \right) \quad (35)$$

References: [33, 25, 10]

6.3 Guo forcing scheme (2002)

In plain words: Body forces (Marangoni, gravity, ...) are added inside LBM collision using Guo's method so the Navier–Stokes force term is recovered.

$$\mathbf{u}^* = \mathbf{u} + \frac{\tau}{\rho} \mathbf{F} \quad (36)$$

$$S_i = w_i \left(1 - \frac{\omega}{2} \right) [3 \mathbf{e}_i \cdot \mathbf{F} + 9 (\mathbf{e}_i \cdot \mathbf{u}) (\mathbf{e}_i \cdot \mathbf{F}) - 3 \mathbf{u} \cdot \mathbf{F}] \quad (37)$$

Lattice force scaling (physical force density $\mathbf{f}_{\text{phys}}$):

$$\mathbf{a}_{\text{lu}} = \frac{\mathbf{f}_{\text{phys}}}{\rho} \frac{\Delta t^2}{\Delta x} \quad (38)$$

References: [18, 25, 3]

6.4 Carman–Kozeny mushy-zone drag

In plain words: In the mushy zone, solid dendrites act like a porous sponge. Velocity is dragged toward zero as $f_l \rightarrow 0$.

$$\mathbf{u}_{\text{final}} = \frac{\mathbf{u}^*}{1 + C_K \frac{(1 - f_l)^2}{f_l^3 + \varepsilon}} \quad (39)$$

References: [40, 32, 3]

7 Free surface (VOF) and surface tension

7.1 VOF phase field

In plain words: $\phi = 1$ means “full of metal”, $\phi = 0$ means gas. The free surface is where ϕ transitions. Tracking ϕ lets the bead crown move.

Volume fraction $\phi \in [0, 1]$; interface where $|\nabla \phi| > \varepsilon$.

References: [19, 35]

7.2 Continuum Surface Force (CSF)

In plain words: Surface tension pulls the interface toward lower curvature (wants a smoother, rounder surface). CSF turns that into a volumetric force near the interface.

$$\hat{\mathbf{n}} = \frac{\nabla \phi}{|\nabla \phi|} \quad (40)$$

$$\kappa = -\nabla \cdot \hat{\mathbf{n}} \quad (41)$$

$$\mathbf{F}_{\text{CSF}} = \gamma \kappa \nabla \phi \quad (42)$$

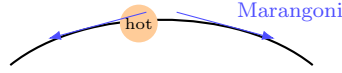
References: [8, 3]

7.3 Marangoni force

In plain words: Surface tension depends on temperature. For many pure metals $d\gamma/dT < 0$: hotter surface has weaker tension, so liquid is pulled toward cooler edges — outward surface flow.

$$\nabla_s T = (\mathbf{I} - \hat{\mathbf{n}}\hat{\mathbf{n}}^\top) \nabla T \quad (43)$$

$$\mathbf{F}_M = \frac{d\gamma}{dT} \nabla_s T |\nabla\phi| \quad (44)$$



Surfactants (S, O) can flip $d\gamma/dT$ and reverse circulation.

References: [24, 12, 30, 3]

7.4 Wetting and contact angle

Young's law:

$$\cos\theta = \frac{\gamma_{sv} - \gamma_{sl}}{\gamma_{lv}} \quad (45)$$

Effective interface normal at wall (Ding–Spelt style):

$$\hat{\mathbf{n}}_{\text{eff}} = \sin\theta \hat{\mathbf{n}}_{\text{wall}} + \cos\theta \hat{\mathbf{t}}_{\text{wall}} \quad (46)$$

Capillary length $\lambda_c = \sqrt{\gamma/(\rho g)}$; Bond number $\text{Bo} = \rho g L^2 / \gamma$.

In the code: `enable_wetting`, `surface_wetting.contact_angle_deg`.

References: [13, 8, 14, 2, 3]

8 Weld-pool body and surface forces

In plain words: After heat melts metal, *forces* shape the pool: surface tension, temperature-driven shear, electromagnetic pinch, gas jet, vapour recoil, gravity. They are cleared once per step, then added with += (never overwriting).

Target force density:

$$\begin{aligned} \mathbf{F}_{\text{total}} = & \gamma \kappa \nabla\phi + \frac{d\gamma}{dT} \nabla_s T |\nabla\phi| + \hat{\mathbf{n}} p_{\text{arc}} + \hat{\mathbf{n}} p_{\text{recoil}} + \boldsymbol{\tau}_{\text{gas}} + \hat{\mathbf{n}} p_{\text{drop}} \\ & + \mathbf{J} \times \mathbf{B} + \rho \mathbf{g} f_l - \rho g \beta (T - T_{\text{ref}}) \hat{\mathbf{z}} f_l \end{aligned} \quad (47)$$

(Sign of Boussinesq: $+z$ upward $\Rightarrow$ hot fluid rises; code uses consistent lattice g .)

8.1 Lorentz (electromagnetic) force

$$\mathbf{F}_{\text{em}} = \mathbf{J} \times \mathbf{B} \quad (48)$$

Potential / Ohm / magnetostatic assembly in conductive metal; gated by `enable_lorentz`.

References: [30, 22, 3]

8.2 Boussinesq buoyancy and hydrostatic gravity

$$F_z^{\text{Buoy}} = -\rho g \beta (T - T_{\text{ref}}) f_l \quad (49)$$

$$\mathbf{F}_g = \rho \mathbf{g} f_l, \quad \mathbf{g} = (0, 0, -g) \quad (50)$$

In the code: `enable_buoyancy`, `enable_hydrostatic_gravity`.

References: [30, 24, 3, 2]

8.3 Arc pressure (Lin–Eagar)

In plain words: The arc also pushes down on the pool surface (like a soft jet). Peak pressure scales with I^2 .

Default model (`arc_physics.pressure_model: lin_eagar`):

$$p_0(I) = \frac{\mu_0 I^2}{4\pi^2 \sigma_p^2}, \quad p_{\text{arc}}(r) = p_0 \exp\left(-\frac{r^2}{2\sigma_p^2}\right) \quad (51)$$

Alternative: constant `pressure_pa`.

In the code: `physics/weld_forces.py lin_eagar_peak_pa`.

References: [30, 39, 3]

8.4 Vapor recoil

Clausius–Clapeyron (preferred when enabled):

$$P_{\text{vap}}(T) = C_{\text{acc}} P_{\text{ref}} \exp\left[\frac{L_v}{R_v} \left(\frac{1}{T_b} - \frac{1}{T}\right)\right] \quad (T \gtrsim T_{\text{onset}}) \quad (52)$$

In the code: `enable_recoil, use_recoil_clausius_clapeyron, recoil_accommodation`. Often off for conduction-mode WAAM.

References: [28, 38, 3]

8.5 Gas-jet shear

$$\tau_{\text{gas}} = \frac{1}{2} \rho_{\text{gas}} v_{\text{jet}}^2 C_{\tau} \quad (53)$$

In the code: `enable_gas_shear, gas_jet_velocity_m_s`.

References: [29, 30, 3]

9 Droplet deposition and impact

9.1 Mass balance

$$m_{\text{dep,expected}} = \dot{m} t, \quad \chi_m = \frac{m_{\text{dep,sim}}}{m_{\text{dep,expected}}} \approx 1 \quad (54)$$

References: [3, 41]

9.2 Impact velocity and pressure

$$L_{\text{drop}} \approx \frac{v_{\text{wire}}}{f_{\text{drop}}} \quad (55)$$

$$v_{\text{grav}} = \sqrt{2g L_{\text{drop}}} \quad (56)$$

$$v_{\text{impact}} \approx f_{\text{mode}} \max(v_{\text{wire}}, v_{\text{grav}} + v_{\text{pinch}}) \quad (57)$$

$$p_{\text{impact}} \approx \frac{\frac{1}{2} m_{\text{drop}} v_{\text{impact}}^2}{\pi r_{\text{contact}}^2} \quad (58)$$

Droplet enthalpy:

$$H_{\text{drop}} = \rho c_p T_{\text{drop}} + \rho L_f, \quad T_{\text{drop}} = T_l + \Delta T_{\text{superheat}} \quad (59)$$

(+ stick-out preheat when enabled).

References: [41, 27, 6, 3]

10 Electrical stick-out and CTWD

$$R_{\text{stick}} = \frac{\rho_e L_{\text{stick}}}{A_w} \quad (60)$$

$$P_{\text{preheat}} = \eta_{\text{stick}} I^2 R_{\text{stick}} \quad (61)$$

$$\Delta T_{\text{preheat}} = \frac{P_{\text{preheat}}}{\dot{m} c_p} \quad (62)$$

CTWD drift with measured bead height:

$$\text{CTWD}_{n+1} = \text{CTWD}_{\text{nominal}} + (h_{\text{bead,measured}} - h_{\text{layer,planned}}) \quad (63)$$

References: [23, 20, 2, 3]

11 Material properties

$$c_{p,\rho} = \rho c_p \quad [\text{J}/(\text{m}^3 \text{K})] \quad (64)$$

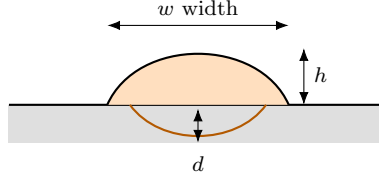
$$L_\rho = \rho L_f \quad [\text{J}/\text{m}^3] \quad (65)$$

$$\gamma(T) \approx \gamma_0 + \frac{d\gamma}{dT} (T - T_{\text{ref}}) \quad (66)$$

Optional piecewise-linear tables in material YAML for $\mu(T)$, $k(T)$, $c_p(T)$.

References: [20, 24, 3]

12 Bead geometry metrics



$$w_{\text{pool}} = \text{transverse liquid extent at surface} \quad (67)$$

$$d_{\text{pool}} = \text{penetration below substrate} \quad (68)$$

$$h_{\text{bead}} = \text{deposit height above substrate} \quad (69)$$

$$\theta_{\text{toe}} \approx \text{contact angle at triple line} \quad (70)$$

References: [3, 24, 2]

13 Coordinate frames (robot integration)

KUKA TCP [mm] $\rightarrow$ simulation metres:

$$x_{\text{sim}} = \frac{x_{\text{TCP,mm}} - x_{\text{origin,mm}}}{1000} - x_{\text{offset,m}} \quad (71)$$

$$y_{\text{sim}} = \frac{y_{\text{TCP,mm}} - y_{\text{origin,mm}}}{1000} - y_{\text{offset,m}} \quad (72)$$

$$z_{\text{sim}} = \frac{z_{\text{TCP,mm}} - z_{\text{origin,mm}}}{1000} \quad (73)$$

Torch z vs CTWD conventions are handled in `coupled_step._resolve_arc_k`.

References: [3, 26]

14 Digital twin platform signals

$$\mathcal{U}(t) = \{t, x, y, z, I, V, \text{WFS}, \text{is_welding}, \text{material_id}\} \quad (74)$$

$$\mathcal{S}(t + \Delta t) = \text{WAAMTwin.step}(\mathcal{U}(t)) \quad (75)$$

Supervisory deviation: $\Delta y = y_{\text{meas}} - y_{\text{twin}}$.

References: [1, 21, 17, 23, 4]

15 Physics tiers (current project)

In plain words: Not every run needs every force. Jobs select a *physics tier* so you can debug thermal behaviour alone, or turn on full weld-pool mechanics.

Tier	Intent	Typical enables
thermal	heat + phase only	no VOF / CSF / Marangoni / MHD
flow	free-surface flow	+ VOF, CSF, Marangoni, buoyancy, gravity, droplets
full	production catalogue	+ Lorentz, Lin–Eagar arc pressure, gas shear, wetting, ...

Set the tier with `simulation.physics_tier` in the job YAML.

Recoil remains an explicit flag (`enable_recoil`); conduction-mode WAAM often leaves it off. Example: `jobs/examples/bead_on_plate_hires.yaml` uses `physics_tier: full` with Goldak + many force flags on.

16 Dimensionless groups

$$\text{Re} = \frac{\rho UL}{\mu} \quad \text{Pr} = \frac{\nu}{\alpha} \quad (76)$$

$$\text{Pe} = \frac{UL}{\alpha} \quad \text{Ma}_{\text{LBM}} \approx \frac{u_{\text{ref}} \Delta t}{\Delta x} \quad (77)$$

$$\text{We} = \frac{\rho U^2 L}{\gamma} \quad \text{Bo} = \frac{\rho g L^2}{\gamma} \quad (78)$$

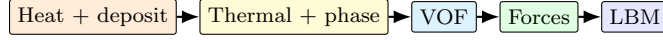
References: [5, 20, 25]

17 Coupled timestep order (current code)

Per-step sequence in `solvers/coupled_step.py` (simplified):

1. Resolve arc location on the free surface / torch z .
2. Arc heat injection (Gaussian or Goldak) → wire droplets → droplet impact pressure/momentum.
3. Thermal advection–diffusion; boundary losses; phase update.
4. Evaporative cooling (optional); enthalpy vapor cap.
5. Optional bead freeze / substrate growth solidification.
6. VOF advection, reinit, wetting BC, flag update.
7. **clear forces**, then add: CSF (+wetting) → Marangoni → gas shear → arc pressure → recoil → hydrostatic gravity → Boussinesq → Lorentz.

8. Force/Mach clamps (stability); LBM collide (Darcy in mush) $\rightarrow$ stream.
9. Tracer advection; thermal-history accumulators; force snapshot.



References: [3, 40, 8, 18]

18 Quick symbol sheet

Symbol	Meaning	SI
I, V	arc current, voltage	A, V
η	arc thermal efficiency	–
Q_w	electrical power in code (IV)	W
Q_{eff}	ηIV useful heat rate	W
$\dot{q}_{\text{arc}}$	local arc heat density	W/m ³
H	volumetric enthalpy	J/m ³
f_l	liquid fraction	–
ϕ	VOF metal fraction	–
a_f, a_r, b, c	Goldak semi-axes	m (mm in YAML)
f_f, f_r	Goldak heat fractions ($f_f + f_r = 2$)	–

19 Bibliography

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